

Commerce XI Maths App

Activities 2.1–2.3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 2 - Functions | Total questions: 8

Question 1 - A2.1 $f \circ g$

If $f(x) = (x + 3)/(x - 2)$ and $g(x) = (2x + 3)/(x - 1)$, find $(f \circ g)(x)$ with its domain.

Options

- A. $x, x \neq 1$
- B. $x, x \neq 2$
- C. $1/x, x \neq 1$
- D. $(3x + 2)/(x - 1)$

Correct answer: $x, x \neq 1$

Explanation and steps

Substitution gives $f[g(x)] = x$. The inner function g is undefined at $x = 1$, so the domain is $\mathbb{R} - \{1\}$.

Question 2 - A2.1 $g \circ f$

If $f(x) = (x + 3)/(x - 2)$ and $g(x) = (2x + 3)/(x - 1)$, find $(g \circ f)(x)$ with its domain.

Options

- A. $x, x \neq 2$
- B. $x, x \neq 1$
- C. $1/x, x \neq 2$
- D. $(2x + 3)/(x - 2)$

Correct answer: $x, x \neq 2$

Explanation and steps

Substitution gives $g[f(x)] = x$. The inner function f is undefined at $x = 2$, so the domain is $\mathbb{R} - \{2\}$.

Question 3 - A2.1 Verification

For $f(x) = (x + 3)/(x - 2)$ and $g(x) = (2x + 3)/(x - 1)$, which conclusion is correct?

Options

- A. $f \circ g \neq g \circ f$ because their domains differ
- B. $f \circ g = g \circ f$ because both simplify to x
- C. $f \circ g = 1$ and $g \circ f = 0$
- D. Neither composite exists

Correct answer: $f \circ g \neq g \circ f$ because their domains differ

Explanation and steps

Both composites simplify to x , but their domains are different: $\mathbb{R} - \{1\}$ and $\mathbb{R} - \{2\}$. Therefore they are not equal as functions.

Question 4 - A2.2 Ordered pairs

For $f(n) = 3n^2 - 4n + 2$, $n \in \{0, 1, 2, 3, 4\}$, select the set of ordered pairs.

Options

- A. $\{(0, 2), (1, 1), (2, 6), (3, 17), (4, 34)\}$
- B. $\{(0, 0), (1, 1), (2, 4), (3, 9), (4, 16)\}$
- C. $\{(0, 2), (1, 1), (2, 4), (3, 11), (4, 22)\}$
- D. $\{(2, 0), (1, 1), (6, 2), (17, 3), (34, 4)\}$

Correct answer: $\{(0, 2), (1, 1), (2, 6), (3, 17), (4, 34)\}$

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives $\{(0, 2), (1, 1), (2, 6), (3, 17), (4, 34)\}$. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 5 - A2.2 Range

For $f(n) = 3n^2 - 4n + 2$, $n \in \{0, 1, 2, 3, 4\}$, determine the range.

Options

- A. $\{1, 2, 6, 17, 34\}$
- B. $\{0, 1, 2, 3, 4\}$
- C. $\{2, 1, 4, 11, 22\}$
- D. $\{2, 6, 17, 34\}$

Correct answer: $\{1, 2, 6, 17, 34\}$

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives $\{1, 2, 6, 17, 34\}$. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 6 - A2.3 Inverse

If $f(x) = 5x - 2$, $x \geq 0$, find $f^{-1}(x)$.

Options

- A. $(x + 2)/5$, $x \geq -2$
- B. $(x - 2)/5$, $x \geq 0$
- C. $5x + 2$, $x \geq -2$
- D. $1/(5x - 2)$, $x \geq 0$

Correct answer: $(x + 2)/5$, $x \geq -2$

Explanation and steps

Let $y = 5x - 2$. Solving for x gives $x = (y + 2)/5$; interchange variables to get $f^{-1}(x) = (x + 2)/5$. Since the range of f is $[-2, \infty)$, the domain of f^{-1} is $x \geq -2$.

Question 7 - A2.3 $f^{-1}(7)$

If $f(x) = 5x - 2$, $x \geq 0$, find $f^{-1}(7)$.

Options

- A. $9/5$
- B. 1
- C. $5/9$
- D. $7/5$

Correct answer: $9/5$

Explanation and steps

$$f^{-1}(7) = (7 + 2)/5 = 9/5.$$

Question 8 - A2.3 $f(x) = 0$

If $f(x) = 5x - 2$, $x \geq 0$, for what value of x is $f(x) = 0$?

Options

- A. $2/5$
- B. $-2/5$
- C. $5/2$
- D. 0

Correct answer: $2/5$

Explanation and steps

Solve $5x - 2 = 0$ to obtain $x = 2/5$.